MUHAMMAD IMRAN

ZOO502 - Animal Physiology and Behavior

Current paper solved question midterm December 2019

1. What is the basic theme in physiology 2? marks

Animal Physiology deals with the study of functions of the tissues, organs and organ systems of animals.

Our major goals of study of animal physiology are to:

explore the physiological processes that are basic to all animal groups

show how they have been shaped by selective forces during evolution

Central themes in animal physiology:

As we study animal physiology and physiological adaptations, we notice several basic themes, repeatedly emerging.

We shall briefly discuss five major themes here, i.e.

(I) Structure-function relationships

(ii) Adaptation, Acclimatization and Acclimation

(iii) Principle of homeostasis

(iv) Feedback control systems

(v) Conformity and Regulation

2. What is the disease caused by hyperthyroidism.2marks?

Excessive secretion of thyroid hormone is known as hyperthyroidism.

The most common form of hyperthyroidism is **Graves'** disease. It is an autoimmune disorder in which antibodies that mimic TSH bind to the receptor for TSH and cause sustained thyroxine production.

- ✓ high body temp.
- ✓ profuse sweating
- ✓ weight loss
- ✓ Irritability
- ✓ high blood pressure
- ✓ protruding eyes (exophthalmia)

3. What is the function of fibers of zonula and ciliary muscels.3marks?

In higher vertebrates, image is focused by changing the curvature and thickness of the lens. The curvature and thickness of the lens is controlled by the suspensory ligaments (fibers of Zonula) and ciliary muscles.

Role of Fibers of Zonula

The shape of the lens is changed by the fibers of the zonula that held the lens in place.

These fibers can exert an outwardly directed tension on the perimeter of the lens.

Role of Ciliary muscles

Ciliary muscles are attached with the fibers of zonula.

Their contraction and relaxation adjust the amount of tension exerted on the lens.

When the ciliary muscles contract, the lens is flattened by elastic tension exerted by the fibers of the zonula, which pull the perimeter of the lens outward. In this state, objects far from the eye are focused on the retina, but objects close by would be fuzzy.

When the ciliary muscles relax, lens becomes more rounded. This focuses the objects close to the eye.

4. Write about the genetic basis of color blindness.3marks

Genetic Basis of Color Blindness

The three types of opsins found in color pigments are encoded by three different genes.

The gene encoding the opsin in blue-absorbing pigment is located on an autosomal chromosome. The genes for red-absorbing and green-absorbing pigments are closely linked on the X chromosome.

Color blindness is caused due to a mutation in one of the cone opsin genes, resulting in absence of one type of pigmented cones.

A person missing a single type of color receptive cones is unable to distinguish some colors.

5. What is meant by range fractionation, also give its importance and give at least two examples.5marks

Definition
In a sense organ, sensory receptors are arranged in an order of increasing sensitivity to different range of intensities of the stimulus. This hierarchical arrangement of receptors is known as range fractionation.
Range Fractionation and Stimulus Intensities
Each individual receptor, in a sense organ, covers only a fraction of the total dynamic range of the stimulus.
Receptors work together in an orderly way to provide discrimination of stimulus intensities.
Recruitment Phenomenon
One important implication of range fractionation is the recruitment phenomenon.
• The most sensitive receptors in the population produce a response at stimulus intensities that are just above the threshold. Above that intensity, the most sensitive receptors become saturated.

• If the stimulus energy is increased a little, the less sensitive receptors in the population will join in and respond.

• With still greater stimulus intensities, another, formerly inactive lower-sensitivity population of receptors will join in.

• In this way, as the stimulus intensity is increased, receptors that are less and less sensitive will become active, until the least sensitive sensory fibers will finally be recruited, and all receptors will respond maximally.

• At that point, the system will be saturated and therefore unable to detect further increases in intensity. This phenomenon of activation of receptors in a graded fashion is called recruitment.
Importance of Range Fractionation
• It results in increasing the active range of multineural sensory system than the range of any individual single receptor.

• It also increases the overall precision of the sense organ.

• It enables the sensory processing centers of the CNS to discriminate stimulus intensities over a range much greater than that of any single sensory receptor.

• The extended dynamic range of the entire system is possible because individual receptors of a sensory system cover different parts of the full spectrum of sensitivity.
Range Fractionation Example:
Eyes have rod cells which are high sensitivity photoreceptors while cone cells have low sensitivity.
At low light intensity, i.e. dimmer light, only rod cells are recruited and we can see dimly, as black/white image. As the brightness increases, cone cells start to recruit and we are able to see colors. Very bright light gives us perception of truly colored image.
After the highest intensity limit, increasing intensity does not increase clarity of the image because the system becomes saturated.

6. Write at least five properties of hormones.5 marks

1.Hormones act only on specific target cells/tissues
2. hormone molecules come into contact with all the tissues in the body during transport through blood, only the cells that contain receptors specific for a particular hormone are affected by it, and generate response.
3.Hormonal action depend on the type of receptor

4.The action of a hormone depends on the nature of enzyme cascade linked to the hormone's receptor, as well as the effector molecules expressed in a particular tissue.

For this reason, a hormone can act on two or more different types of tissues and generate different types of responses.
For example, Antidiuretic hormone stimulates water reabsorption in the nephron of kidney while the same hormone produces vasoconstriction and increases blood pressure by acting on the walls of arteries. The receptors in kidney nephron tubule and arterial receptors are different, causing the same hormone to generate different responses.
5. Hormones are active at very low concentration

The amount of hormone produced by an endocrine gland is small. It is further diluted in the blood and interstitial fluid. The available concentration for the target cell lies between 10-8 to 10-12 M and the hormone is still effective at this very low concentration.
The high sensitivity of hormonal signaling is due to the high affinity of target cell receptors for hormones.
6. Hormonal effects are amplified

Binding of a hormone molecule to its receptor leads to a cascade of enzymatic steps that amplify the effect; thus just a few hormone molecules can influence thousands or millions of molecular reactions within a cell.

7.Write note on Acetylcholine 5 marks

Acetylcholine
• The most familiar fast acting neurotransmitter substance is the Acetylcholine.
• In most instances, acetylcholine has an excitatory effect.
• It also acts as inhibitory transmitter in some instances e.g. at the peripheral parasympathetic nerve endings where it is involved in the inhibition of the heart by the vagus nerves.
Cholinergic Neurons
Neurons that release Acetylcholine are said to be cholinergic. They are widely distributed throughout the animal kingdom. These neurons include:
• vertebrate motor neurons
• preganglionic neurons of the vertebrate autonomic nervous system
• postganglionic neurons of the parasympathetic division of the autonomic nervous system
• many neurons of the vertebrate central nervous system
• a number of invertebrate neurons e.g. cells of the molluscan central nervous system, motor neurons of annelid worms, and sensory neurons of arthropods.
Mode of Action of Acetylcholine
• When acetylcholine (ACh) is released into the synaptic cleft, it binds to the ligand-gated ACh-specific receptors in the postsynaptic membrane.
• The binding causes Na+ and K+ ion channels to open briefly that produce an EPSP.

8.Photoreceptors 3 marks

Photoreceptors:
Photoreceptors possess light-sensitive pigments which are carotenoids retinal and 3-dehydroretinal associated with opsin proteins to form rhodopsins.
These pigments absorb photons of light energy and then produce a generator potential.

External energy, such as light, may strike any part of the body; but only the eyes contain sensory cells that can receive and respond to the stimulus of light and transduce photons into neuronal energy.

9.Basic type of stimuli.2marks

types of Stimuli
Two types:
1. **External stimuli** :Stimuli of odor, touch, light, sound and gravitation
2. **Internal stimuli** : Stimuli of pain, homeostatic imbalances and blood pressure

10.Stigma and ocelli 5 marks

Eyespot or Stigma
The simplest photoreceptive structures are the Eyespots or Stigma found in some protozoa e.g. euglena. Stigma is a bright red colored structure having carotenoid pigments. It gives a sense of light and dark and helps in phototaxis.
Eyecups or Ocelli
Cnidarians and flatworms have multicellular photoreceptive structures call eyecups or Ocelli. They consist of a cuplike depression containing photoreceptor cells. They cannot form image and only provide the animal a sense of direction.

11.Deuteronatope and paranatope 3 marks.

Protanope and Deuteranope

A person with loss of red cones is called a protanope.
In this type, the overall visual spectrum is shortened at the long wavelength end because of a lack of the red cones.
A color-blind person who lacks green cones is called a deuteranope.
Such a person has a normal visual spectral range because red cones are available to detect the long wavelength red color.

12.Donnan equilibrium 3marks

Discovery
In 1911 Frederick Donnan discovered the phenomenon of unequal distribution of ions across the two sides of a differentially permeable membrane. This phenomenon is known as Donnan equilibrium and can be defined as:
Definition

> "If diffusible solutes are separated by a membrane that is freely permeable to water and electrolytes but totally impermeable to one species of ion, the diffusible solutes become unequally distributed between the two compartments".

13.Role of corpus luteum in pregnancy.

Role of Corpus Luteum in Pregnancy

If the released ovum is fertilized and becomes implanted in the endometrium, the developing placenta begins to produce chorionic gonadotropin hormone that maintains an active corpus luteum, so that estrogen and progesterone secretion continues until the placenta fully takes over the production of these hormones, at which time the corpus luteum degenerates.
In many other mammals, such as the rat, the corpus luteum, stimulated by prolactin, continues to grow and secrete estrogen and progesterone throughout the gestation period.

14.Explain negative feedback with 2 example.5

The secretory activities of most endocrine tissues, especially those involved in maintaining homeostasis, are modulated by negative feedback.
In this type of feedback, the hormone itself or products resulting from its action tend to suppress its further release.
Negative feedback mechanisms ensure a proper level of hormone activity at the target tissue and prevent its over-activity by over-secretion.
Negative feedback is a type of regulation in biological systems in which the end product of a process in turn reduces the stimulus of that same process. Feedback, in general, is a regulatory mechanism present in many biological reactions. By allowing certain pathways to be turned off and on, the body can control various aspects of its internal environment. This is similar to flipping a switch. Feedback allows the product of a pathway to control the switch. Sometimes referred to as a "negative feedback loop", negative feedback occurs when the product of a pathway turns the biochemical pathway off
Negative feedback systems:
In life, the most common form of regulation encountered is the negative feedback, in which accumulation of an end product works to stop or slow down that process.
Example-1
The breakdown of sugar in the cells generates chemical energy in the form of ATP. When a cell makes more ATP than it can use, the excess ATP "feeds back" and inhibits an enzyme near the beginning of the pathway. This results in temporary stoppage of ATP production.
The figure illustrates the negative feedback:
The three-step chemical pathway shown here converts substance A to substance D. A specific enzyme catalyzes each chemical reaction. Accumulation of the final product (D) inhibits the first enzyme in the sequence, thus slowing down production of more D.
Example-2
The control of blood sugar (glucose) by insulin is another good example of a negative feedback mechanism.

> When blood sugar rises, receptors in the body sense a change. In turn, the control center (pancreas) secretes insulin into the blood effectively lowering blood sugar levels. Once blood sugar levels reach homeostasis, the pancreas stops releasing insulin.

15.Abnormaities due to growth hormones.3marks

Developmental Abnormalities Due to GH

Disturbances in the secretion of growth hormone lead to several patterns of abnormal growth and development in humans:
Gigantism: excessive size and stature caused by hypersecretion of growth hormone during childhood (before puberty).

Acromegaly: enlargement of the bones of the head and of the extremities caused by hypersecretion of growth hormone beginning after maturity.
Dwarfism: abnormal underdevelopment of the body caused by insufficient secretion of growth hormone during childhood and adolescence.

16.Androgen hormone secreted by adrenal cortex.2marks

Hormones of Adrenal Cortex

The adrenal cortex produces two major types of steroid hormones: the mineralocorticoids and the glucocorticoids collectively called as corticosteroids. These hormones are involved in blood ion and glucose regulation and anti-inflammatory reactions.

- Adrenal cortex also secretes small amounts of sex hormones, especially androgenic hormones.
- These androgens exhibit about the same effects in the body as the male sex hormone testosterone.

17.Give sliding filament theory, 5 marks

Sliding Filament Theory

H. E. Huxley and A. F. Huxley proposed the sliding filament theory of muscle contraction in 1954.
This theory states that during muscle contraction the thin and thick filaments in sarcomeres slide and undergo shifting.
When a muscle contracts, the thin actin filaments actively slide along between the thick myosin filaments and move closer to the center of the sarcomere. As a result, the sarcomere becomes shorter.
When a muscle relaxes or is stretched, the overlap between thin and thick filaments is reduced, and the sarcomere elongates.
The changes in sarcomere length during contraction and stretch of a muscle, correspond to changes in muscle length.

Explanation

• In a relaxed muscle fiber, the thick and thin filaments overlap only at the ends of A band. But when muscle fibers are stimulated by the nervous system, myosin heads are attached to the myosin-binding sites on actin in the thin filaments, i.e. cross bridges are formed and the sliding begins.

head

• During contraction, the A bands (myosin filaments) maintain a constant length, whereas the I bands and the H zone (zones where actin and myosin filaments do not overlap) become shorter and Z lines get closer.

• When the muscle is stretched, the A band again maintains a constant length, but the I bands and H zone become longer.

• Neither the myosin thick filaments nor the actin thin filaments change their lengths when a sarcomere shortens or is stretched. It is the extent of overlap between actin and myosin filaments that changes.

Length-Tension Curve

One of the strongest pieces of evidence in support of the sliding filament theory comes from the length-tension relation of a sarcomere.
Experimental measurement of the shortening of length of sarcomere during contraction and resulting force generates a length-tension curve. This curve explains the assumptions of sliding-filament theory.

Explanation of the Curve

• The tension produced by the muscle is maximal when the overlap between thick and thin filaments allows the largest number of cross-bridges to be formed between actin and myosin.

• Tension drops off with increased length of sarcomere, because the thick and thin filaments overlap less and fewer crossbridges can be formed.

• It also drops off with decreased length, because thin filaments begin to collide with one another, preventing further shortening.

• The curve also predicts the consequence of sliding filament theory that no active tension will develop if a sarcomere is stretched so far that there remains no overlap between actin and myosin filaments, making it impossible to develop any crossbidges.

Conclusion

This curve shows that the tension produced by a sarcomere is proportional to its shortening which is due to sliding of thick and thin filaments and formation of cross bridges in the sarcomere during contraction. These were the proposals of sliding filament theory. So length-tension curve provides a practical proof of this theory.

18.What are smooth muscle and there general function?.

Smooth Muscles

General Features

• Least specialized muscle fibers.
• Have myosin similar to that found in contractile nonmuscle cells.
• Non-striated.
• Involuntary: under autonomic control.
• Contract and relax slowly.
• Capable of more sustained contractions.

19.Teste buds and types of cell teste buds.

Taste buds

• The gustatory organs of vertebrates are called taste buds.

The taste bud is composed of about 50 modified epithelial cells including supporting cells (sustentacular cells), basal cells and taste receptor cells.

• The basal cells are progenitor cells that give rise to new taste receptors. They regularly generate new sensory taste receptor cells which have an active life of only 10 days.

Taste Receptor Cells

• The outer tips of the taste cells are arranged around a minute taste pore.

• From the tip of each taste cell, several microvilli, or taste hairs, protrude outward into the taste pore to approach the cavity of the mouth.

• These microvilli provide the receptor surface for taste.

• Interwoven around the bodies of the taste cells is a branching terminal network of taste nerve fibers that are stimulated by the taste receptor cells.

20. Sensory adaptation with example.

Definition

The decrease in perceived intensity, when the intensity of the stimulus has not itself changed, is known as sensory adaptation. The phenomenon is due to reduction in the frequency of sensory response during a sustained stimulus.

Adaptation Example:

Wearing clothing stimulates touch receptors at all points where our garments touch the skin. But we typically adapt to the touch input from our garments. So we can easily detect any new touch stimuli that impinge on our skin, even at locations covered by our clothing.
Other common examples of adaptation under our everyday observation are with the stimuli of smell, hot or cold water and dim or bright intensity of light.

More Examples

• Stimuli of smell
• Hot or cold water
• Dim or bright light

21.Feedback mechanism principle and significance.

Feedback Control Systems:

Definition:
Many biological processes have the ability of self-regulation by a mechanism which is called feedback mechanism. The basic principle of feedback in living systems is that the output or product of a process itself regulates the process.

Significance:
The feedback regulatory processes maintain homeostasis in the cells and the body of multicellular organism as a whole.

Mechanism: Feedback controls regulate particular variables e.g.,

- Temperature
- Salinity
- pH

This regulation requires:

- Continuous sampling of controlled variables
- Respective corrective actions.

22.What is homeostasis. 2marks

Homeostasis—definition:
During evolution, each species has assumed a specific set of internal environment with an ability to resist environmental changes by making adjustments to keep its internal fluctuations in a narrow range. This ability to protect internal environment from the harms of fluctuations in external environment is termed as homeostasis. Homeostasis does not mean to keep a fixed internal environment as the changes maintained within a specific range are necessary for normal body functions.

Example:

1. Homeostatic regulation of water
2. Homeostatic regulation of temperature
3. Regulation of pH
4. Regulation of glucose concentration
5. Regulation of osmotic pressure
6. Regulation of oxygen level
7. Regulation of ion conc.

23.Phasic or tonic function .

Adaptation: "The reduction in the frequency of action potentials that is typically seen in a neuron during asustained stimulus". Two types:

- Phasic response
- Tonic response

Adaptation: Phasic response: The type of adaptation in which neurons adapt rapidly & stop generating more APs after initial 1 or 2 APs.

Adaptation: Tonic Response: The adaptation in which neurons adapt slowly & fire repetitively to a subthreshold stimulus, but with gradually decreasing frequency.

24.Calcium ion role in excitation.

Transmission of AP through T tubules results in the release of Ca^{2+} ions form stores of sarcoplasmic reticulum. This increases Ca^{2+} conc. inside the muscle fiber in the immediate vicinity of myofibrils.

- These calcium ions cause the contraction to begin. So, the net effect of excitation-contraction coupling is to link an AP in the plasma membrane of the muscle fiber to the concentration of free Ca^{2+} in the cytosol that initiate contraction.

25.Name different component of membranous labyrinth.2 marks

Membranous Labyrinth:

It consists of three parts:

- Three semicircular canals
- Vestibule (sacculus and utricles)
- Cochlea

- The semicircular canals and vestibule are concerned with maintaining balance while cochlea is the sensory structure concerned with hearing.

26.What is reason behind latency period in action potential.

Latency Period in Coupling

It takes several milliseconds to begin contraction after the arrival of an AP.
This latency is because of the large size of skeletal muscle fibers which cannot contract unless action potential spreads deep into the fiber to the vicinity of each myofibril.
During this latent period, action potential is transmitted along the transverse tubules (T tubules) deep within the fiber.

27. What are the effect of neuromodulation.3

Effects of Neuromodulation:

The phenomenon of neuromodulation plays critical roles in the development of complex behavioral patterns e.g. happiness, revenge, reward, greed and exploration. Neuromodulation is also involved in the processes of thinking, cognition, planning, learning and memory.
Many behavioral problems are also based on neuromodulation e.g. mood swings, sleep disturbances, feelings of stress, anxiety, depression and anger.

28.Name different component involve in the mechanism of conductivity of heart.5 marks

The atrial internodal pathways that conduct impulse from the S-A node to the atrioventricular (A-V) node.

• The A-V node, in which the impulse from the atria is delayed before passing into the ventricles.

• The A-V bundle, which conducts the impulse from the atria into the ventricles.

• The left and right bundle branches of A-V bundle.

• Purkinje fibers, which branch off from bundle branches and conduct the cardiac impulse to all parts of the ventricles.

29.What type of cells are present in islet of lengerhens?

Islets of Langerhans

• The islets contain four types of cells:

1. Beta Cells

• constitute 60% of the Islets. They secrete insulin

2. Alpha Cells

• constitute 25% of the islets .They secrete glucagon

3. Delta Cells

• constitute 10% of islets. Secrete somatostatin that inhibits the secretion of insulin and glucagon.

4. PP Cells

• Present in small number. Secrete a hormone of uncertain function called pancreatic polypeptide.

30.Function of estrogen in pregnancy? 5

Functions of Estrogens in Pregnancy

During pregnancy, estrogens exert mainly a proliferative function on most reproductive and associated organs of the mother. They cause:
(1) enlargement of mother's uterus
(2) enlargement of mother's breasts and growth of breast ductal structure
(3) enlargement of mother's external genitalia.
They also relax the pelvic ligaments of the mother, so that the pubic symphysis becomes elastic. These changes allow easier passage of the fetus through the birth canal.

31.Diff b/w cGMP and cAMP? 3

Differences b/w cGMP and cAMP Pathways
A few of such differences include:

• The production of cGMP is catalyzed by guanylate cyclase enzyme from GTP.

• Guanylate cyclase occurs in two forms: one bound to the plasma membrane and one free in the cytoplasm. In contrast, adenylate cyclase is always bound to the plasma membrane.

• Guanylate cyclase becomes active as the Ca^{2+} concentration is increased within the cell, while adenylate cyclase activity is increased when Ca^{2+} conc. is low.

• cGMP activates a specific protein kinase, protein kinase G instead of protein kinase A.

32. Pace marker and it's types.

Pacemaker
The pacemaker is the excitatory region which generates the rhythmical impulses that control the rhythmicity of cardiac chambers.
Pacemaker Types
Two basic types of pacemakers in animals with pumping hearts:
• **Neurogenic pacemakers**

• **Myogenic Pacemakers**

Neurogenic and Myogenic Pacemakers

A. In many invertebrate hearts, the pacemaker is neurogenic that consists of neurons. These hearts are known as neurogenic hearts.

B. The pacemaker in some invertebrate and all vertebrate hearts is myogenic i.e. consists of specialized self-excitatory muscle cells. Such hearts are known as myogenic hearts.

33.Gap junction. Writes its works

Gap Junctions

Electrical activity, initiated in the pacemaker region of the heart, spreads over the entire heart from one cell to another because the cells are electrically coupled via membrane gap junctions.

Gap junctions are regions of low resistance between cells and allow current flow from one cell to the next across intercalated disks.

Role of Gap Junctions

• Allow current flow from one cell to the next.

• Pacemaker's electrical activity spreads over the entire heart via gap junctions.

34. Note on axon.

Axon

• An axon is a long and thick process that arises from the cell body and has a constant diameter.

• The lengths of the axons vary from few millimeters to more than a meter.

• The longest axons are seen in whales, where the axon of a single spinal motor neuron may extend many meters from the base of the spine to the muscles in the tail fin.

Axon—Functions

• The axons are specialized to conduct signals away from the cell body.

They have evolved mechanisms that allow them to carry information for long distances with high fidelity and without loss.

Structural components of axons:

The axon contains a jelly-like semi-fluid substance called Axoplasm, surrounded by the plasma membrane which is called Axolemma.

Axon has mitochondria and ER but lacks Nissl's granules. So exons are not involved in protein synthesis.

Axon terminals:

• Each axon may divide into numerous branches at its termination. These branches of axon are called axon terminals.

• Each axon terminal has small extensions called telodendria with enlarged ends called terminal knobs. These knobs have granules or vesicles that contain neurotransmitters.

• The axon terminals allow transmission of signals to many other neurons, glands or muscle fibers simultaneously.

Axon hillock:

• The region of an axon where it joins the cell body is cone-shaped and is called the axon hillock.

• This is the region where the signals that travel down the axon are generated.

Myelin sheath:

• Specialized neuroglial cells, known as Schwann cells are located at regular intervals along the axons of many neurons.

• These Schwann cells secrete a fatty layer known as myelin sheath, over the axons.

• The myelin sheath is not continuous, but there are non-myelinated points in between, which are called Nodes of Ranvier.

35.Mammalian Ear decision.

The Mammalian Ear

Ear

• Ear is the organ of hearing that detects sound within a particular frequency range.

• The human ear can detect sound frequencies lying between 20 to 20000 hertz.

Functional Anatomy of Mammalian Ear

waves through three principally different media i.e., air, bone and fluid.

• The conducted waves are converted into electrical signals through a series of complex steps.

• Mammalian ear has 3 major divisions:

• External ear

• Middle ear

• Inner ear

The 3 parts are designed to conduct sound

External Ear: External ear includes:

• The pinna (auricle)

• External auditory meatus
• Tympanic membrane (eardrum) .
These structures act as a funnel to collect sound waves from environment
Sound is amplified and concentrated onto the eardrum that vibrates

Middle Ear

• Contains three auditory ossicles malleus, incus, stapes in a series.
• Malleus attached to eardrum and stapes to oval window of cochlea.

Functions of Middle Ear

1. Receive sound waves as vibrations from eardrum and transmit them onto oval window of cochlea.
    1. Amplify sound waves (more than 20 fold).

Inner Ear

• It consists of a bony labyrinth which is filled with perilymph and has an oval and a round window.
• Oval window receives sound vibrations through
stapes.
• In the bony labyrinth lies the membranous labyrinth that is filled with endolymph.

36.Myelin sheath function.

Myelin Sheath

• Myelin sheath is a thick and multi-layered membranous structure that surrounds many vertebrate axons.

• It is mainly composed of a fatty substance sphingomyelin.

Functions of Myelin Sheath:

I. Electrical insulation
II. Increase the speed of conduction
III. Space efficiency
IV. Saltatory conduction

Electrical Insulation:

• The sheath mainly contains lipid substance sphingomyelin that is a poor conductor of electrical current.
• It provides electrical insulation for the axon.
• It decreases ion flow about 5000-fold.
• So, it acts as an analogue to the plastic insulation
around electrical wires.

Increase the Speed of Conduction:

• Insulation with myelin enhances the efficiency of longitudinal spread of current.
• So, it results in rapid transmission of nerve impulse.
• It renders thin axons higher conduction speed than very thick non-myelinated axons.
• e.g. 20 μm myelinated axon has faster speed than 40x thicker squid giant axon.

Space Efficiency:

• Thousands of myelinated axons can be packed into space occupied by just one giant axon.

Saltatory Conduction:

• Ions cannot flow through internodes that have thick myelin sheath, so APs are not generated in these regions.
• Ions flow only through the nodes of Ranvier.
• Inward Na+ current produced at a node during the rising phase of AP travels all the way to the next node.
• Here it depolarizes the membrane and regenerates the AP.
• AP appears to jump along the axon from node to node.
• This mechanism of conduction is called saltatory conduction.

37. sarcoplasmic reticulum (5marks).....

Sarcoplasmic Reticulum

• The sarcoplasmic reticulum (SR) is a network of membrane-bound tubules that extends

throughout muscle cells on either side of a Z disk and extends from one Z disk to the next as well.

• In many features, it is similar to the endoplasmic reticulum in other cells.

• The SR has a special organization that is extremely important in controlling muscle contraction.

• That is why the rapidly contracting types of muscle fibers have extensive network of sarcoplasmic reticulum.

38.sarcolemma and sarcoplasmic

The membrane of the cell is called sarcolemma, its cytoplasm is called sarcoplasm and its endoplasmic reticulum is known as sarcoplasmic reticulum.

39.Define cardiac and stroke volume ?2

Cardiac Output

Cardiac output is the quantity of blood pumped into the aorta each minute by the heart. This is also the quantity of blood that flows through the circulation.

The average cardiac output for the resting adult is often stated to be almost exactly 5 L/min.

Factors Affecting Cardiac Output

Cardiac output varies widely with the level of activity of the body.

The following factors directly affect cardiac output:

(1) Basic level of body metabolism

(2) Physical activity of the body

(3) Age

(4) size of the body

Stroke Volume

The volume of blood ejected by each beat of the heart is termed the stroke volume.

Stroke volume is the difference between the volume of the ventricle just before contraction (end-diastolic volume) and the volume of the ventricle at the end of a contraction (end-systolic volume).

40.Give the name hormone of adrenal cortex 3

A. Aldosterone
B. Cortisol,
C. corticosterone
D. cortisone

41.Exocrine glands (5 marks)

Exocrine glands

Exocrine glands produce fluid secretions that are delivered through ducts onto the epithelial surfaces of the body. Exocrine glands are more easily identified than endocrine glands because of their duct leading to the body surface. The fluid secretions may be either proteins (enzymes) or mucous or both.

Examples:

• Salivary glands produce saliva that is delivered to the oral cavity for partial digestion of food through parotid and submandibular ducts.

• Pancreas produces enzyme-containing pancreatic juice that is delivered to the small intestine through pancreatic duct.

• Lacrimal glands produce tears that is delivered through lacrimal duct on the surface of eye to provide lubrication.

• Mammary glands produce milk that is delivered through lactiferous ducts to the nipples for nourishing the young.

1. The main function of endocrine glands is to secrete hormones directly into the bloodstream
2. The Pineal Gland Runs a Daily Biological Clock.
3. The Thyroid and Parathyroid Glands Increase Metabolism and Regulate Calcium Levels.

42.hrmons secreted by ovaries

I. Estradiol(estrogen)
II. Progesterone

43.uses of kymograph 2 marks

Kymograph

A kymograph is a device that graphically records changes in the mechanical activities of animal tissues in the physiological experiments.

Uses of Kymograph

Since its invention in the 1840s, the kymograph has been used most commonly in the field of medicine to study various physiological and muscular processes.

It has been used to study the skeletal muscle contractions (twitch and tetanus) as well as the cardiac muscle activities (cardiac cycle).

It is also used to measure blood pressure and rate of respiration.

44.three main parts of ear 2 marks

Mammalian ear has 3 major divisions:

1. External ear
2. Middle ear
3. Inner ear

• The 3 parts are designed to conduct sound waves through three principally different media i.e., air, bone and fluid.

45.Sensory organs of humans.2 marks.

The major sensory organs of human body are eyes, ears, nose, tongue and skin.

46.Positive and negative feedback mechanism.3marks.

Positive feedback systems:

There are many biological processes that are regulated by positive feedback, although they are less commonly found. In positive feedback systems, an end product speeds up its production by enhancing the effect of original stimulus.

The figure explains the positive feedback system in a biochemical pathway. A product stimulates an enzyme in the reaction sequence, increasing the rate of production of the product.

Negative feedback systems:

In life, the most common form of regulation encountered is the negative feedback, in which accumulation of an end product works to stop or slow down that process.

47.Myosin molecules structure and functions 5 marks

Myosin Molecule

• Each myosin molecule consists of two identical heavy chains which are coiled together to form a long tail. It also has two globular heads which are made from two heavy chains plus three or four calcium-binding light chains.

• The heads form cross bridges between the thick and the thin myofilaments during contraction.

Thin Filaments

• The thin filaments are 7-8 nm thick and extend across the I band.

• They are composed chiefly of actin molecules.

• Thin filaments also overlap myosin filaments in the peripheral darker regions of the A band. In these regions, six actin filaments surround each myosin filament while each actin filament is surrounded by three myosin filaments.

• In thin filaments, actin molecules are arranged in two chains which twist around each other.

• Two strands of another protein tropomyosin twist around the actin and help to stiffen it. In a relaxed muscle fiber, they block myosin binding so that the myosin heads can not bind to the thin filaments.

• Thin filaments also have a three polypeptide complex troponin at intervals of about 40nm along the thin filament.

• One of the troponin polypeptides (TnI) is an inhibitory subunit that binds to actin, other (TnT) binds to tropomyosin and helps position it on actin while third (TnC) binds the calcium ions.

• Both troponin and tropomyosin help control the myosin-actin interactions involved in contractions.

The H Zone

The center of A band appears lighter than the other regions in a relaxed sarcomere.

• This region is called H zone and contains only thick filaments. There are no overlaps between the actin and myosin in this region.

• The H zone is bisected by a dark line, the M line which contains enzymes important in energy metabolism.

48.ca ion role in excitation .

Hair cell excitation due to inward K^+ current also causes the opening of voltage-gated calcium channels.

The resultant Ca^{2+} influx causes transmitter (mainly glutamate) release from the basal end of the cell onto the auditory nerve endings, stimulating them to send an electrical signal along the cochlear nerve.

49.Three growth hormons.

1. Piturity

2. Placenta
3. hypothalamus

50.Two vascular resistance

Vascular resistance is the resistance that must be overcome to push blood through the circulatory system and create flow.

- systemic vascular resistance (SVR
- total peripheral resistance (TPR)

51.What is myosin how it helps in muscular contraction

• When a muscle contracts, the thin actin filaments slide b/w the thick myosin filaments & move closer to the center of sarcomere. As a result, sarcomere becomes shorter.

52.Group of hormones insoluble in lipids.3 marks

The second messengers involved in signal transduction fall into three distinct groups:
• Cyclic nucleotide monophosphates e.g. cAMP (cyclic adenosine monophosphate) and cGMP (cyclic guanosine monophosphate).
• Inositol phospholipids e.g. inositol trisphosphate ($InsP_3$) and diacylglycerol (DAG)
• Ca^{2+} ions and associated calmodulin

53.hair cilia

cilia (plural), are small hair-like protuberances on the outside of eukaryotic cells. They are primarily responsible for locomotion, either of the cell itself or of fluids on the cell surface.

54.Difference Pacinian corpuscles and meissners corpuscles.

Meissner's Corpuscles: are the touch receptors. They have encapsulated nerve endings which lie in papillae which extend into the ridges of fingertips.
Sense of touch perceived through skin having Meissner's corpuscles
Pacinian corpuscles: are situated quite deep in the skin. They have encapsulated nerve endings and receive deep pressure stimulus.
Pacinian corpuscles that are situated deep in the body and receive pressure stimulus. In the limbs, they receive vibrations.

55.myofilament and its types 3markss

Each myofibril is composed of myofilaments. Myofilaments are of two types, thin filaments and thick filaments.
Thin filaments are composed of actin while the thick filaments are composed of myosin molecules.

56.What is meant by sensation?

Sensations arise when sensory receptor cells, receiving a stimulus, transmit signals through the nervous system to particular parts of the brain that interprets or perceives these signals. The subjective description of these neuronal perceptions is termed as sensation.